



Himanshu Vaidya

Founder, CTO and Technology Head, Visionary Business and Technology Leader, AI Expert, Quantum SME, Distinguished Research Scientist, Pioneering Global Leader in establishing Centers of Excellence in Emerging and Next-Gen Software Technologies

Researcher, Business Leader, and Technology Expert with Unique Leadership Experience driven through Innovation, Discipline, and Passion (Company Website: www.eqspanse.com; Personal Website: <https://himanshuvaidya.in>)



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BRIEF SUMMARY

Niche Expertise as a Technology Evangelist for Scientific/R&D Software Systems and Research Laboratories (04/2002 - Present)

As a Quantum AI Cybersecurity expert in India, I have consulted the MOD, GOI, for incorporating Quantum and AI Technology based specifications in India's first Drone Framework published by MOD for Indigenous Manufacturing of Drone Tech in India. I possess proven experience in developing and implementing data and AI software on Robotics and Informatics equipments for industrial R&D setups and academic research laboratories - I have worked in US at TDCC (Dow) Global R&D HQ at Midland, MI. I have also managed data from various scientific instruments in fields such as Astronomy & Astrophysics, Cosmology, Material Science, Chemistry, Synoptic Meteorology and Climatology, Geoscience/Earth Science, and Space Technology, including Atmospheric Remote Sensing. I worked with world-class research institutions on international research projects to obtain this data, such as IUCAA, NCRA, TIFR, ORT, RAC, TIFR, IIG, IMD, IITM, ISRO, IRSC, AWS AI and Quantum, Xanadu Cloud, IBM Q-Experience, Fraunhofer Research Institute, Karlsruhe Institute of Technology (KIT), Cassidian (Airbus Defence and Space Systems, formerly EADS), Saarland University, etc. Developed and deployed Data Acquisition and Analytics Software on 2 International Radio Telescopes: The Giant Meterwave Radio Telescope (GMRT), and The Ooty Radio Telescope (ORT), facilities of The Tata Institute of Fundamental Research, India. Worked on software systems for the Geoscience Laboratories of an Oil and Gas Giant and implemented High Throughput Research Informatics and Robotics software for Dow Chemical Company's core R&D laboratory infrastructure. This included working on equipment for Formulation Science and catalysis Synthesis, which used chromatography principles, Raman spectroscopy, etc. Photometry and Spectroscopy instruments are also used for several of my research projects implemented in Astrophysics and Cosmology, where data of stellar spectra, Sloan Digital Sky Survey, and Dark Energy Spectroscopic Survey are used for photometric and spectroscopic redshift observations of galaxies and quasars. Application of emerging and next-generation technologies in the Quantum, Data, AI, Cloud and IOT tech space has been my forte, which I have used to implement application software around instrumental setups for Industrial Automation, Industrial IoT, Pharmaceutical and Medical R&D, Healthcare, Automotive, Aerospace, Defense, and Oil and Gas and Mining Business Domains.

Proven Technology and Business Leader in establishing Emerging and Next Generation Technology Global Centres of Excellence (COE/GCC) from scratch (04/2002 - Present)

Established 3 COEs for 3 different employers: 1. For Capgemini in the field of Engineering Analytics and Edge Machine Learning, 2. For Globant in the Field of Big Data, Artificial Intelligence, and IoT, and 3. For Altimetrik in the field of Data Engineering, Data Science, Generative AI, and Quantum Computing. Proven track record of strategic management in the field of emerging and next-generation technology. I have established myself as a Technology Visionary and Business Scientist who can implement technology solutions catering to specific businesses, from requirements analysis to product deployment. Mentored Oxford University MBA Graduates' established Space Tech Startups for developing edge Machine Learning software on the Satellite Edge processors - a project funded by Delloite and collaborated with AWS Space Tech Cloud experts, under the Gravity 3 challenge. I established myself as a pioneering leader in Quantum Computing and Quantum AI in the state of Maharashtra and volunteered for Quantum and AI Education in the past 7 years. For this work, I received a National Award, "Excellence and Innovation in AI Education" at the Global Artificial Intelligence Symposium and Awards 2021, under the Niti Ayog, Government of India, from the All India Council for Robotics and Automation. Selected by AWS, sponsored by the Ministry of Electronics and Telecommunication (MeitY), Government of India, for a 24-month R&D project in Quantum AI Algorithms for Astrophysics and Cosmology, for which I am collaborating with Global Leaders such as Quantum Hardware OEMs like Xanadu, IBM, and global quantum teams of AWS and Xanadu.



CERTIFICATES

Finance for Non-Financial Leaders (02/2019 - Present)

ISRO's Earth Observation Data Hub - Bhoonidhi (01/2025 - Present)

Fundamentals of Deep Learning from NVIDIA Deep Learning Institute (DLI) (11/2021 - Present)

Building Transformer Based NLP Applications from NVIDIA DLI (11/2021 - Present)

Fundamentals of Accelerated Data Science with RAPIDS from NVIDIA DLI (03/2022 - Present)

Applications of AI for Anomaly Detection from NVIDIA DLI (03/2022 - Present)

Deep Learning on AWS from Amazon (11/2020 - Present)

Practical Data Science with AWS Sagemaker from Amazon (09/2020 - Present)

High Throughput Research Informatics and Robotics BootCamp at TDCC R&D HQ in Midland, MI, USA (09/2007 - Present)

VB.NET, ASP.NET, WPF, WCF from Microsoft Corporation (09/2007 - Present)



WORK EXPERIENCE

Founder, CTO and Head of Technology (Indian Deep Tech Startup in Quantum AI Software for AeroSpace, DefSpace, & Space Tech) eQspanse NextGen Technologies Private Limited

04/2024 - Present

Pune, India

As a Quantum AI organization, we specialize in harnessing emerging and disruptive technologies like Generative/Agentic AI + Quantum to drive innovation in Cybersecurity, Cryptography, Industrial Automation, and Autonomous Systems. Our focus lies in building cutting-edge platforms and delivering bespoke services and solutions that leverage the power of these Next Generation Technologies for Predictive Maintenance (Diagnostics and Prognostics), System Simulations, and Autonomous Systems of Unmanned Vehicle (UV) / Drone + Satellite Tech. By integrating Quantum technology and Artificial Intelligence, we elevate traditional approaches, unlocking new possibilities and accelerating progress to build customized software products and solutions. With a commitment to pushing boundaries and exploring the forefront of technological advancements, we empower businesses to thrive in a rapidly evolving digital landscape. We build research-led innovative software platforms and solutions for multiple industry verticals (Aerospace, Defense, Space Tech, Deep Tech, Automotive, Manufacturing, Healthcare, Pharmaceuticals, Medical Research, Retail / Supply Chain Consumer Goods, Banking + Fintech), with current focus on the DefSpace segment. We identify the specific Industry problem domains and use cases that can only be solved using a combination of Quantum and Artificial Intelligence technologies, packaged with Data, IOT, and similar allied groups of technology stacks to deliver an end-to-end solution to the industry problems of our clients. We apply Evolutionary + Intelligent Algorithms based on AI Agents, Agentic + Gen AI, Multi-Agent Systems + Swarm Intelligence based AI / Quantum tech, Quantum Cryptography + Cybersecurity protocols + algorithms, Optimization tools + simulations to solve industry use cases, like: geospatial surveillance, space situational awareness, near-earth threat identification for planetary defense, systems simulation, precision medicine, drug discovery, predictive maintenance, equipment/asset diagnostics and prognostics, and intelligent cyber-physical systems for drone cluster / satellite cluster communication, simulation and intelligence (COMINT, ELINT). For this, We provision our generic quantum-enhanced software platform, and meticulously craft it to provide bespoke technology solutions for the underlying problem industry domain/use case of choice.

Achievements/Tasks

- Contributed as a Quantum, AI, and Cybersecurity SME to consult and design the Drone Framework published by MOD, GOI for Cybersecurity of Drone Manufacturing
- Heading Innovation and Technology through R&D and Product (Platform) Engineering. Our software utilizes state-of-the-art solutions in Gen AI / Agentic AI, Multi-Agent Systems, Quantum AI Cybersecurity (PQC/QKD), SLM, LLM, vLLM, MCP, A2A and AP2.
- Established collaborations with the commercial Wing of ISRO - InSpace, Indian Remote Sensing Centre (IRSC), Ministry of Electronics and Telecommunications (Meity), GOI, Ministry of Defence (MOD), GOI.
- Connected with Quantum Hardware OEMs across the globe (Xanadu) and Cloud Service Provider Hyperscalers like IBM, NVIDIA, AWS, Xanadu and Microsoft for using their Quantum Hardware for running our niche algorithms in Quantum and AI end-to-end tech stack
- Mentored Oxford Incubated Space Tech Startup and lead technology solutions for deploying Edge AI Algorithms on Satellite Processor for the same. This was part of the Gravity03 Global Challenge, sponsored by Deloitte and partnered with Amazon. I used AWS Sagemaker, AWS Ground-station, and other AI services on Amazon Cloud.
- Designed Faculty Development Programs (FDP) under the ATAL Innovation Fund from the Ministry of Education, and taught comprehensive hands-on sessions of multiple days to 100+ Faculties and Professors across India
- Working on multiple research and academic collaborations for Quantum Technologies and AI-related cutting-edge innovation collaborations within India and International Research Communities, from EU, Amazon, Xanadu, IBM, and Google
- Designed UGC-approved Quantum Computing and Quantum AI courses for B.E./B.Tech. students, also taught the same from past 3-4 years
- Establishing Tie-ups with multiple Universities and Engineering Colleges in India and abroad for Quantum AI Research and Education (e.g. the Netherlands Consulate in Mumbai, Denmark Technical University, QuTech, etc.)
- Pioneered as a Leader in Quantum AI Education and Research in India, and extensively worked in this space in the state of Maharashtra and across India (Atmanirbhar Bharat)

Contact : Monika Himanshu - 9011994555



CONFERENCES & COURSES

IEEE Distinguished Visitor (09/2025 - Present)

3rd IEEE DVP EXA Global Event

- Key Note Speaker for a Global IEEE Conference. Talked on "Industrial Applications of Quantum and AI Technologies".

Global Artificial Intelligence Summit and Awards (09/2021 - 09/2021)

All India Council for Robotics and Automation, New Delhi, India

- Participated as a Delegate in the Summit. Received the award "Excellence and Innovation in AI Education" for the ongoing Ph.D. Research Work in Quantum AI and Education Initiatives in Quantum Computing and Quantum AI (<https://youtu.be/dguNCA39jbq>)

Pune Data Conference 2019 (Largest Data Conference in Pune City, India) (03/2019 - 03/2019)

Clairvoyant

- Participated as a Speaker in Big Data and AI (<https://www.youtube.com/watch?v=w9PVIwpZlXy>)

Designed and Taught University Courses in Quantum Computing (2021 - Present)

Vishwakarma Institute of Information Technology

- Designed 2 UGC (University Grants Commission of India) approved courses in Quantum Computing: i) 6 months elective in Quantum Computing, and ii) Dual Degree course with Honors in Quantum Computing for B.E./B.Tech. students



WORK EXPERIENCE

Principal Investigator [Part-Time] (24 Months Quantum AI Research Project)

Funded by Ministry of Electronics and Information Technology,
Selected by AWS

05/2023 - 04/2025

Pune, India

I am working as the Principal Investigator (PI) for this research proposal in Quantum Computing.

Achievements/Tasks

- Received approval for Cohort 2 Call for Proposals from Ministry of Electronics and Information technology (Meity), Government of India, Quantum Computing Applications Laboratory (QCAL)
- Received a research grant solely in the form of AWS Credits worth USD 160000 for the project titled: "State-of-the-art Hybrid Quantum-Classical AI Algorithms for Astrophysics and Cosmology", with a validity for next 24 months as a Research Project in Quantum Computing
- We are aiming to develop novel Hybrid Quantum AI Algorithms involving Deep Quantum Learning, Hybrid Quantum Deep Neural Networks, AWS Bracket, Xanadu Cloud, Cosmology Datasets from NASA like Sloan Digital Sky Survey and Dark Energy Spectroscopic Survey
- Applied mesh-TensorFlow for model parallelism. Used NVIDIA libraries, tools and services like RIVA services, NEMO, TAO toolkit. Applied multiple large language models for Deep Learning and Big Data Machine Learning. Familiar with Few-shot/Zero-shot learning using large pre-trained models as a state-of-the-art industry practice for NLP/NLU and Computer Vision.
- Utilizing State-of-the-art Deep Neural Network and Hybrid Quantum-Classical Deep Neural Networks for Patterns Recognition on Graphs, Manifolds and Meshes. We are implementing Deep Quantum Convolutional Neural Networks (CNN) like Graph CNN, Anisotropic CNN, Manifold CNN, and Geodesic CNN.
- Using datasets from NASA and ESA, especially from the Sloan Digital Sky Survey (SDSS), and Dark Energy Spectroscopic Survey (DESS), which provides galaxy catalogs containing photometric and spectroscopic redshifts of galaxies and quasars.
- Explored specific Algorithms related to Quantum-Enhanced Geometric and Topological Deep Learning, Quantum-Inspired Machine Learning, and Hybrid Quantum-Classical AI Algorithms applied to Astrophysics, Cosmology, Space Tech, and Deep Tech
- Proven expertise established in State-of-the-art Research in the Interdisciplinary and multidisciplinary field of Quantum AI
- Identified Pros and Cons of different Quantum Hardware Systems available and tailored Quantum Software Algorithms accordingly for advanced problems in the field of Astrophysics and Cosmology
- Identified industry application areas for state-of-the-art algorithms discovered and implemented during this ongoing research



VOLUNTEER EXPERIENCE

Member of Industrial Advisory Board, Visiting Faculty and Resource Person for FDP (Quantum Computing, AI)

Vishwakarma Institute of Information Technology

03/2020 - 02/2025

Pune, India

Deemed University with Engineering and Multidisciplinary Courses

Tasks/Achievements

- Contributed as an Advisory to The Industry Advisory Board for Computer Science and AI/Data Science
- Designed Syllabi (teaching as a Visiting Faculty) for 2 courses in Quantum Computing, approved by The University Grants Commission
- Conducted multiple sessions as a Resource Person in Faculty Development Programs for Emerging Trends in Quantum Computing and Quantum AI

Dr. Vivek Deshpande, Director, VIIT -
Available on Request

Member of AWS Customer Council

Amazon Web Services

05/2021 - 04/2025

Tasks/Achievements

- Voluntary membership received from AWS, where I provide feedback to AWS on their Cloud Computing, AI/ML and Quantum Computing services.

Distinguished Member of Advisory Board and Resource Person for FDP (AI Program)

Ramrao Adhik Institute of Technology, D Y Patil Deemed University

01/2023 - 12/2023

Navi Mumbai

Tasks/Achievements

- Industry Advisory Consultant on AI, Quantum Computing and Big Data



WORK EXPERIENCE

Engineering Leader and Distinguished Architect (Product Engineering, Big Data Analytics and Artificial Intelligence)

Altimetrik

02/2021 - 04/2024

Pune, India

Service Based Giant with Product Development Sister Organization

Achievements/Tasks

- Established myself as a Thought Leader, leading multiple accounts and portfolios across Product and Platform Engineering, Data Architecture and Artificial Intelligence
- Managed team sizes close to 300 in size and business of close to 12M+ USD, overall 30M USD
- Leading Multiple Product and Platform Engineering org-wide initiatives, including innovation in Big Data and AI/ML space
- Participated in Money 2020 international conference related to FinTech specifically supporting the Quantum Computing Applications showcase for Finance Industry
- Conducted and led multiple org-wide trainings and sessions on Quantum, AI and Product Engineering
- Grown new business, expanded existing business focused on Product Engineering, Data and AI for multiple Business Domains (primarily FinTech, Online Retail and Healthcare)
- Leading multiple practices related to Data Engineering, Data Science and Data Analytics: i) Deep Learning, ii) Machine Learning, iii) Generative AI, iii) Conversational AI, iv) Discriminative AI and v) Rule Based AI
- Deployed Solutions related to Intelligent Chatbots, Intelligent Document Processing Engines, Intelligent Optical Character Recognition Solutions and Anomaly Detection Engines for Fortune 100 MNC clientele in Banking & FinTech, Pharmaceuticals & Healthcare, and Space Technology Industries
- Large models used for transfer learning: i) Speech To Text: Jasper, Quartznet, Citrinet (256, 384) ii) Text To Speech: Fastpitch, Waveglow, HiFiGAN iii) Natural Language Processing (NLP): Domain Specific NER, BERT NER, BERT Text Classification, BERT and Megatron QA, BERT Intent and Slot, BERT Punctuation
- Development of AI/ML solutions for Satellite Data Processing for Earth Observations

Technical Director & India Head (Big Data and AI CoE)

Globant

10/2018 - 02/2021

Pune, India

Service Based MNC with focus on Emerging Technologies

Achievements/Tasks

- Established the India division of Big Data and AI CoEs, by working as an Intrapreneur by owning Business, Technology and Talent Responsibilities
- Worked as a **Chief Big Data and AI Architect** to establish Business in Big Data and AI across the EMEA Region, by heading all technology practices in Big Data and AI: i) Data Engineering, ii) Data Architecture, iii) Business Intelligence (Data Integration and Data Visualization), iv) Machine Learning, v) Deep Learning, vi) Conversational AI, vii) Natural Language Processing (NLP), viii) Natural Language Understanding (NLU) and ix) Computer Vision
- Managed P&L for Big Data and AI India CoEs, and established Globant India as a brand in Big Data and AI tech
- Developed and Deployed: i) end-to-end AI solutions for Pharmaceutical Giant client in Intelligent Chatbot and Voice Recognition, ii) video analysis and document processing solutions as part of in-house product for Human Resource Function, iii) image analysis solutions for intelligent detection of psoriasis disease for pharmaceutical giant client
- Established Globant India as a brand in Big Data and AI in India and EMEA region, by sponsoring multiple conferences, workshops and events in AI, Machine Learning and Big Data. Also participated as a Speaker in these events, details of which can be found on my LinkedIn profile



ORGANIZATIONS

All India Council for Robotics and Automation (10/2021 - 09/2022)

Professional Fellow Member of AICRA

Amazon Web Services
(05/2024 - 04/2025)

Member of AWS Customer Council, Providing Advisory Feedbacks on AI and Quantum Computing Services available on Amazon Cloud



SKILLS

AI Agents, Agentic AI, Multi Agent Systems, Intelligent Systems

Agentic AI Protocols: MCP, A2A

MCP Servers, MCP Clients, MCP Design Patterns

LangChain, LlamaIndex, OpenLLM, LangGraph, LangSmith, LangFlow, LangFuse, LangServe,

Quantum and AI Cybersecurity Protocols, AI Based Cryptanalysis

Generative AI Tools: Claude AI, Claude Code, Gemini, ChatGPT Go, GPT-4

Small Language Models (Phi-3, Mistral, TiniLLaMA), Large Language Models (LLM), vLLM, RAG based pipelines, Multi Agent RAG

Prompt Engineering, RAG, Fine Tuning, Transfer Learning, LLM Architectures (GPT, MoE, LRM, VLM, SLM, LAM, HLM, LCM) and Pipelines

Rule Based AI, Conversational AI, Generative AI, Discriminative AI, Agent-Based AI

Diagnostics Analytics, Predictive Analytics, Prescriptive Analytics, Cognitive Analytics

Large Language Models (LLM), Large Vision Models (LVM), Large Language and Vision Assistant (LLaVA), LLaMA 2, YOLO V7, chatGPT, GPT-3, GPT-4, Google Bard, BERT based Architectures like Vision Transformers, Hybrid Models, DALL-E, CLIP, VPT, Shape Google, etc.



WORK EXPERIENCE

Principal Architect & Associate Director (Enterprise Data and AI Architecture, IoT, Engineering Analytics, Edge ML)

CapGemini

04/2016 - 10/2018

Achievements/Tasks

- Worked on multiple IPs in Engineering Analytics, Edge ML, and Mesh Computing
- Developed Copyrighted Software Platform from inception to deployment that utilized various facets of AI and Big Data for Predictive Maintenance using Digital Twins for multiple asset classes used in Mining, Oil, and Gas, and potentially Aerospace Industries
- Developed Mesh Computing Algorithms, Hybrid Architectures (Centralised, Distributed and De-Centralised topology for device clusters in a Edge-Fog-Cloud Industrial Automation system) for co-R&D laboratory involving industrial IOT experimental setup.
- Developed and Deployed an Engineering Analytics Platform from scratch
- Worked extensively on predictive maintenance for Mining, Oil and Gas, and Healthcare and developed solutions for asset management and asset prognostics using Machine Learning and cloud technologies. This involved real-time and batch processing of data from sensors mounted on suites of equipment providing the asset data for specific business use.
- Established Best Practices in Advanced Analytics, Platform Engineering and Co-R&D initiatives with teams working for Fortune 500 MNC Clientele
- Specialized Work in Predictive Maintenance related Analytics Software Development
- Specialized Work in Reusable Components based Library generation on Containerized Cloud Infrastructures (AWS)
- Specialized Work in Enterprise Data Architecture, Pre-Sales Solution Architecture

Senior Engineering Lead (AI Product/Platform Development)

Acellere (Embold, now BrowserStack)

04/2014 - 03/2016

Achievements/Tasks

- Provided Specialized Software and Analytics Consultancy to Fortune 100 MNC clientele in Oil and Gas. for their US (Texas) and China (Beijing) Geoscience Laboratory Teams developing data-intensive software using VC++ and C#.NET.
- Refactored Legacy Code Bases using innovative techniques: executing TDD, Refactoring to improve the design of code, Design Refactoring, Refactoring to Patterns, Dependency Analysis, Call Graph Analysis, Trace flow analysis of program execution traces, overall significantly addressed technical debt for multiple software systems.
- Mentored and Trained team members on overall Acellere Analytics platform, it's implementation, code review, ML applications and Product Engineering in general.
- Data Visualization software for 2D and 3D code visualizations (heirarchical heatmaps and treemaps, 3D cluster graphs and CityView graphs, etc.)
- Predictive analytics software using Machine Learning for identification of change-proneness, bug-proneness, design and architecture of software systems, modules and components



SKILLS

Quantum Deep Convolutional Neural Networks (CNN), Deep Quantum Graph CNN, Deep Quantum Anisotropic CNN, Deep Manifold CNN, Deep Geodesic CNN, Deep Quantum Learning, Geometric Deep Learning, Quantum SVM, Gaussian Boson Sampling framework and associated algorithms on Photonic Quantum Hardware, Variational Quantum Eigen solvers, Graph Kernel Methods, Quantum-inspired AI Algorithms, Quantum Neural Networks, Physics-based Neural Networks

LSTM, RNN, BERT-variants, Decision Trees, Random Forest, XgBoost, CatBoost, AdaBoost, Ensemble methods, Bagging and Boosting ML techniques, Support Vector Machines (SVM), Quantum SVMs, K-Nearest Neighbors, K-means clustering, Graph Clustering, Watershed Algorithm, Fourier and Wavelet Transforms

Geometry, Topology, Homogeneity, Isotropy of 3D and higher dimensional Euclidean and Non-Euclidean spaces, Spherical Harmonics, Point Clouds, Alpha Spaces, Betti Numbers, Cubic and B-Splines, Euler Characteristics, Minkowski Functionals, Quantum Voronoi Tessellations, Delaunay Triangulation

IP Generation (copyrights, trademarks, patents)

Software Product and Platform Engineering, GTM Strategy, Product Rollout

Enterprise Data Architecture, Enterprise AI Architecture, Enterprise Application Architecture

Big Data Machine Learning, Machine Learning with Limited Data (Few Shot, Zero Shot, One-Shot)

Quantum Algorithms, AI Quantum, Quantum Software, Quantum Neural Networks



WORK EXPERIENCE

Senior Technical Specialist (Data & AI Solution Architect) Persistent Systems

08/2013 - 03/2014

Pune, India

Achievements/Tasks

- Developed and deployed a large-scale data migration framework using Java, Multithreading, SpringBatch, and Apache Akka frameworks
- Lead and participated in the initiatives for Product and Platform Engineering, Data and Artificial Intelligence related to Astronomical Data Processing for the Square Kilometer Array (SKA) International project [Virtual Observatory (VO) Project] by collaborating with Inter University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

Engineering Lead / Product Development Specialist Acellere [Embold, now BrowserStack]

08/2011 - 07/2013

Achievements/Tasks

- Worked on developing a Code Analytics Platform and Product from scratch, involving Machine Learning, AI, and Java, Python and Perl languages: Integrated capability to analyze Objective-C language into the Code Analytics Platform from scratch
- Lead multiple International R&D projects funded by the German Government: i) Automatic Identification, Extraction, and Reuse of Software Components (<http://www.kmu-devise.de/>), and ii) Semi-Automatic Analysis of Concurrency Issues in Software Systems. (International Research Collaborations with: Fraunhofer Research Institute, Karlsruhe Institute of Technology (KIT), Saarland University, FZI, etc.)
- Consulted an EU Aerospace, Defense, and Space Tech Giant Fortune 100 MNC (Cassidian, wing of Airbus Defence and Space Systems, formerly EADS) and developed software coding, design and architectural guidelines for their mission-critical systems software

Project Lead / Delivery Manager United Software Associated Private Limited

01/2011 - 06/2011

Achievements/Tasks

- Roles: Project Management, Delivery Management, Client Relationship Management and New Business Development.
- Domain of Projects: a) Live Website Development & Maintenance. b) n-tier web-based distributed application development for ERP/MRP systems in construction and irrigation domain. c) Team Management (9 developers, 1 tester, 5 projects).

Freelanced Consultancy in Software Sadaphal Technnovations

03/2009 - 12/2010

Achievements/Tasks

- Worked as Software Development Consultant for developing ERP software for Construction Industry.
- Published Research Paper on Software Applications using Modeling and Simulations in NCMS 2009 at DIAT (DU), DRDO, Pune.



SKILLS

Quantum Cybersecurity Solutions and Algorithms [NIST Standards]: Post Quantum Cryptography (PQC), Quantum Key Distribution (QKD)

Edge Machine Learning

Engineering Analytics

Advanced Analytics

Data Mesh Architecture

Distributed Application Architecture

Empirical Software Engineering

ML Ops, Data Ops, AI Ops

SDLC, Scrum of Scrum, Kanban

P&L, People, Technology, Program & Revenue Management

Business Expansion, New Business Development

Thought Leadership

Space Sciences, Astrophysics and Cosmology

Mathematical Modeling

Scientific Software Development

MATLAB, R, Python, Java, C, C++, C#, VC++, Perl

PyTorch, Tensorflow, Keras, numpy, scipy, pandas, statsmodels, PyCaret, AutoML

Google Colab, Jupyter Notebook, PyCharm, Spyder, Visual Studio Code, CLion, Eclipse, IntelliJ

Linux, Windows, Mac OS, Raspbian (Raspberry Pi OS)

AWS Sagemaker, AWS and Azure Cloud Services, NVIDIA Deep Learning Cloud Services



WORK EXPERIENCE

Senior Research Computing Specialist (Laboratory Intelligence, Analytics and Visualization)

The Dow Chemical Company (TDCC)

02/2007 - 01/2009

Achievements/Tasks

- Worked on Software Applications in .NET and Java languages for the Informatics and Robotics Laboratories of The Dow Chemical Company
- Specialized Work Experience at Midland, MI, USA and Houston, Texas, USA on High Throughput Research Informatics (HTRi) and Robotics systems
- Worked on Data Mining, Data Visualization and Statistical Pattern Recognition Software Development for multiple laboratory workflows and research projects
- Developed application software on multi-million dollar laboratory instruments in the robotics and informatics laboratories of the core R&D division of Dow
- Analyzed data such as spectroscopic graphs, experimental design setup software, setting up automation protocol for laboratory run on robotics equipments, high throughput research informatics for material synthesis and formulation science, material modeling and simulation software for polyurethane (PU) and PU foams, etc.

Software Engineer

Incotec Software India Private Limited

04/2006 - 12/2006

Achievements/Tasks

- Owned and Implemented a project for converting a VC++ Software Product compatible with Asian Languages. During this migration, analysed 3-tier architecture for Front-End, Middle Layers and Backend of the system for Oracle and SQL Server Backends.
- Worked as a Software Engineer for development and maintenance of a MRP II (part of larger ERP systems) related software product (used for shop-floor scheduling) using VC++, MFC.

Project Trainee / Research Trainee

National Centre for Radio Astrophysics, Tata Institute of Fundamental Research

04/2002 - 07/2005

Pune, India

Research Laboratory Centre governing Two International Radio Telescope Facilities of Government of India

Achievements/Tasks

- Developed and deployed Data acquisition and analytics Software on 2 International Radio Telescopes [Giant Metrewave Radio Telescope (GMRT) and Ooty Radio Telescope (ORT), TIFR]
- Developed software systems for Noise Removal and Signal to noise ratio improvement for real-time data acquisition of telescope data
- Implemented two mathematical models that used 2D Spatial Fourier Transforms and 2D Convolution, for image analysis for detecting/predicting 2D structure of radio galaxies and quasars [Quasi Steller Objects (QSOs)] (452 QSOs observed and modelled for structure estimates for 2D structures like Double Gaussian and Core Halo)
- Implemented and deployed a Data Visualization system for Wind Tomography over the GMRT Array
- First Research Scientist to develop and deploy software system for Interplanetary Scintillation (IPS) observations from GMRT, TIFR, India - long-term reverse application of IPS was to model the Solar System Environment for Space Weather Monitoring applications by accurate modeling of Solar Wind [Interplanetary Medium (IPM)] using ground-based observation techniques from The Earth



SUPPORTED CAUSES

Quantum Computing Awareness

Science and Technology Popularization

AI and Quantum Education

Industry-Academia Collaboration

Collaborative Multi-disciplinary, Cross-disciplinary Research

Research-led Innovation

Student Potential, Youth

Entrepreneurship

Mother Earth



INTERESTS

Science and Technology

Yoga and Meditation

Spiritual Awakening

Music and Harmony, Music Science

Chess, Swimming, Table Tennis, Football

Travelling, Sky Watching, Astronomy

Nature, Gardening

Reading Scientific and Spiritual Books



LANGUAGES

Marathi

Native or Bilingual Proficiency

Hindi

Native or Bilingual Proficiency

English

Full Professional Proficiency

Sanskrit

Limited Working Proficiency



EDUCATION

Ph.D. in Quantum AI applied to Astrophysics and Cosmology

Christ University, Pune, India

06/2021 - Present

Courses

- State-of-The-Art Hybrid Quantum-Classical Algorithms for Detecting Large Scale Structures in The Cosmic Web Data from NASA and ESA Datasets (SDSS/DESS)
- Pursuing multiple courses in Quantum Computing and Artificial Intelligence from NVIDIA, IBM, AWS, etc. (will be shared on request)

M.Sc.(Space Sciences), specialization: Astronomy & Astrophysics

Department of Space Sciences, University of Pune, India

2002 - 2005

Courses

- 3.5 years of Research Work in Radio Astronomy. Title: Structure of Extragalactic Radio Sources using Interplanetary Scintillations (IPS) M.Sc. Thesis submitted to TIFR, Pune Library.
- Acquisition and Analytics Software Developed and Deployed on 2 International Radio Telescopes (GMRT and ORT, TIFR)

Post Graduate Diploma in Advanced Computing (DAC)

Centre for Development in Advanced Computing, Pune, India

2005 - 2006

Courses

- Wind Tomography over The GMRT Array (Java based Visualization Solution for providing real time insights and alarms for studying the wind profile over The Giant Meterwave Radio Telescope)

Bachelor of Computer Science (BCS)

S. P. College, University of Pune, India

1999 - 2002

Courses

- Data Analysis Software Project in COBOL for Processing Meteor Shower Data



HONORS / AWARDS

Excellence and Innovation in AI Education (11/2021 - Present)

All India Council for Robotics and Automation

- Award received in the Global Artificial Intelligence Summit and Awards (GAISA) 2021 as "Excellence and Innovation in AI Education" for voluntary Education and Research work in Quantum Computing and AI
- This symposium was conducted by All India Council for Robotics and Automation and sponsored by The Niti Ayog, Government of India

Multiple Awards in Chess and Music (01/1991 - Present)

National Level Chess Player, Performed and won many music competitions in Natya Sangeet, Classical Music

First Prize in Pre-Military Training Camp (05/1997 - Present)

Hanuman Vuayam Prasarak Mandal, Amaravati, Maharashtra



SOFT SKILLS

Speaker, Performer, Innovator, Scientist, Leader (Leadership by Example)

Chess Player, Drawing, Painting

Music Composition, Vocal + Instrumental Indian Classical Music

Traveling, Nature Lover, Sky Watching



VIDEO SESSIONS / TALKS

Quantum AI Fundamentals

International Association of Business Analytics Certifications (IABAC)

Session Link

- <https://www.youtube.com/live/1nRFgPOxjV4?feature=shared>

Industrial Applications of Quantum [AI] Technologies

IEEE CS DVP 2025 Global event

Session Link

- <https://fb.watch/CG16FAk2Zz/>

Quantum AI Session on World Pi Day

Pimri-Chinchward University

Session Link

- <https://youtu.be/NUQNTqheXak?si=TDI8GkniKqNCrRCV>

ATAL FDP Sessions: (ATAL Innvation Fund)

RAIT AI Program, D Y Patil University, Navi Mumbai

Sesssion Links

- Day 1 Session 1: Topic: "Quantum Computing Fundamentals":
<https://youtu.be/wsPIWc4BQXE?si=SKKYin6ugWbOT8aR>
- Day 1 Session 2: Topic: "Introduction to Quantum Programming":
<https://youtu.be/10Y9whK7hXk?si=ALBzzxPOTTh8103E>
- Day 2 Session 1: Topic: "Quantum Machine Learning Deep Dive":
<https://youtu.be/8ZQoL2Xz71I?si=W8mU756hSyzGFWq->
- Day 2 Session 2: Topic: "Business Applications of Quantum Computing and Quantum Machine Learning (QML)":
https://youtu.be/xPkn_j_DofY?si=zS1Y3shqpYpLB7m0

Session on Open Standards for Big Data and Artificial Intelligence

Pune Data Conference 2019

Session Link

- <https://youtu.be/w9PVLwpZIxY?si=rODf48TDBcTx8HI2>